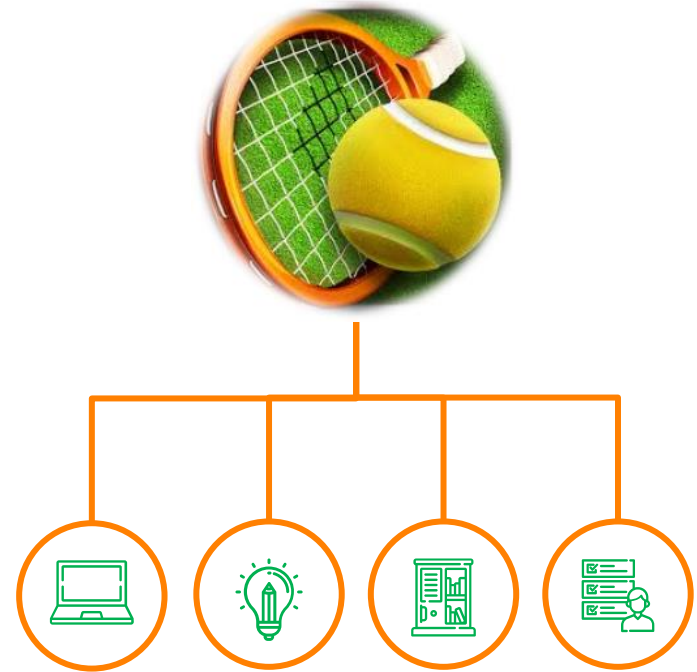


# Tennis Match Data Analysis

Team Members: Hamed Arjmandi –  
Dina Ebrahimnadjari – Diba Wakil Azad

May 2026



# Dataset Overview

## Event

- Winner Code
- Start Datetime

## Power

- Set Number
- Game Number

## Tournament

- Tournament Category
- Ground Type

Match ID



## Home/Away Team

- Player ID
- Full Name
- Current Rank
  - Height
  - Country
- Turned Pro Year

## Statistics

- Statistics Category: service, points, games, return
- Statistics Name: aces, double faults

## Summary Box



**16,873 matches**



**2,637 players**



**101 countries**



**Jan and Feb 2024 data**

# Project Workflow

**Dataset:** Data of professional tennis matches from January and February 2024

## 2. Cleaning Data

After creating the DataFrames, null values and unnecessary columns were dropped. The cleaned data was retained.

## 4. Visualization

Some analysed features were visualized, using Matplotlib.



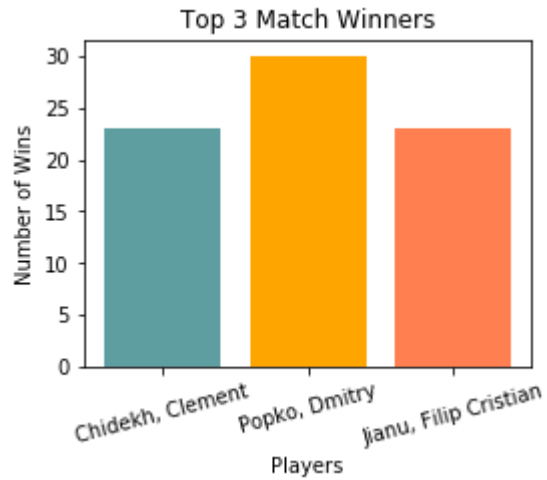
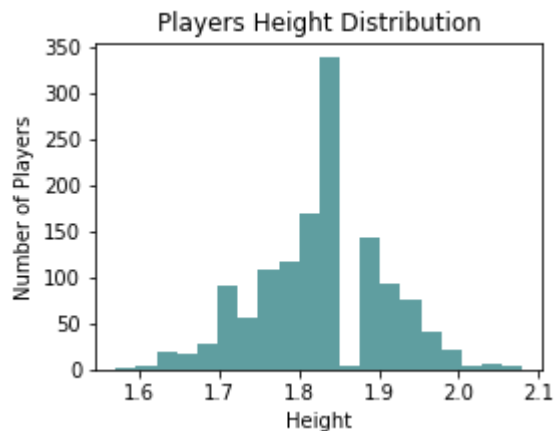
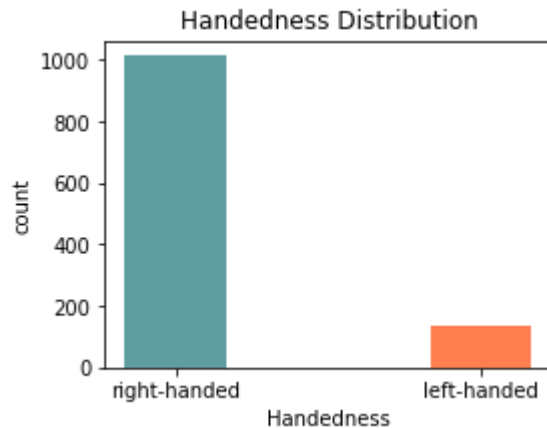
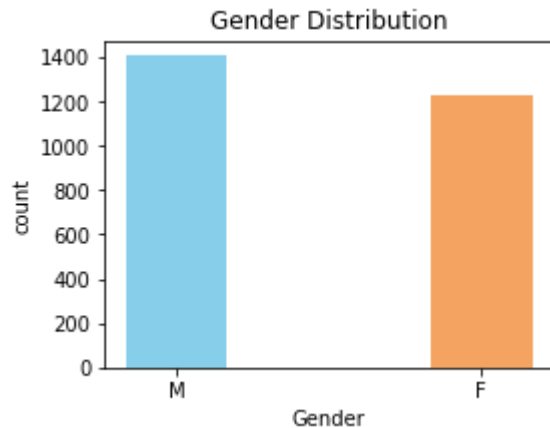
## 1. Loading Data

We read data from parquet files after extracting nested ZIP files. Then concatenated files to create the tables, using pandas.

## 3. Analysing Data

Using the cleaned data, we answered some questions on: player attributes, match dynamics, performance metrics, and environmental factors.

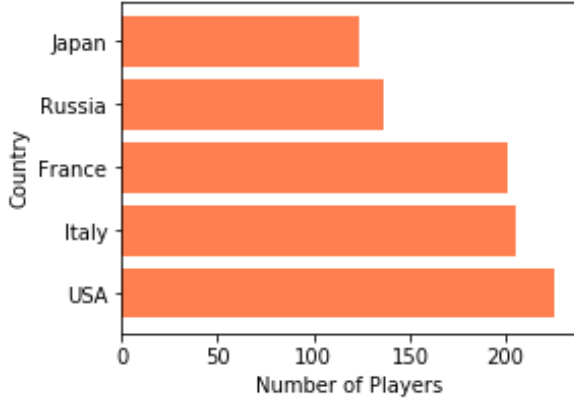
# Players Demographics



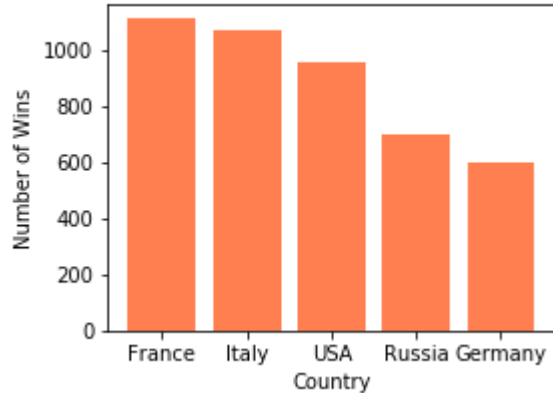
- **1,407 male/1,227 female** (6 missing values)
- Most players are **right-handed**
- Average Height of Players: **1.82 m**
- Most winning player is **Dmitry Popko** from **Kazakhstan, 29 wins**

# Tournament Insights

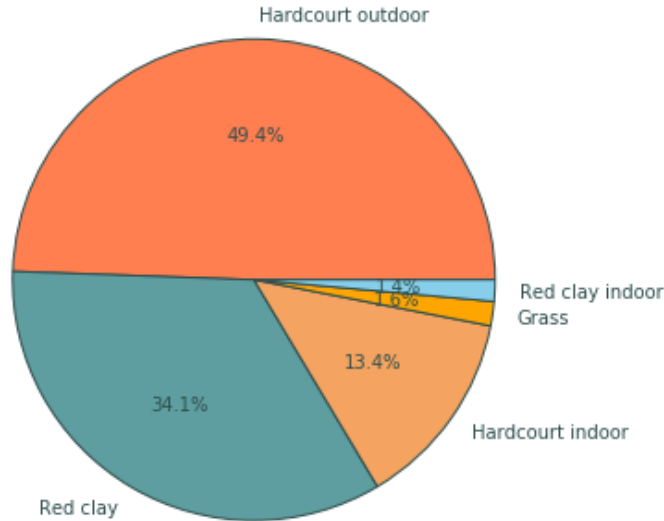
Top 5 Countries By Number of Players



Top 5 Countries by Number of Wins

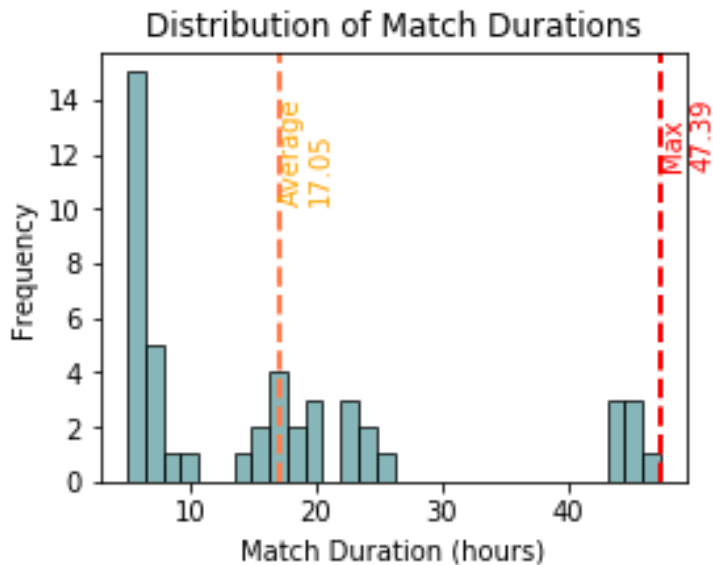


Distribution of Match Surfaces



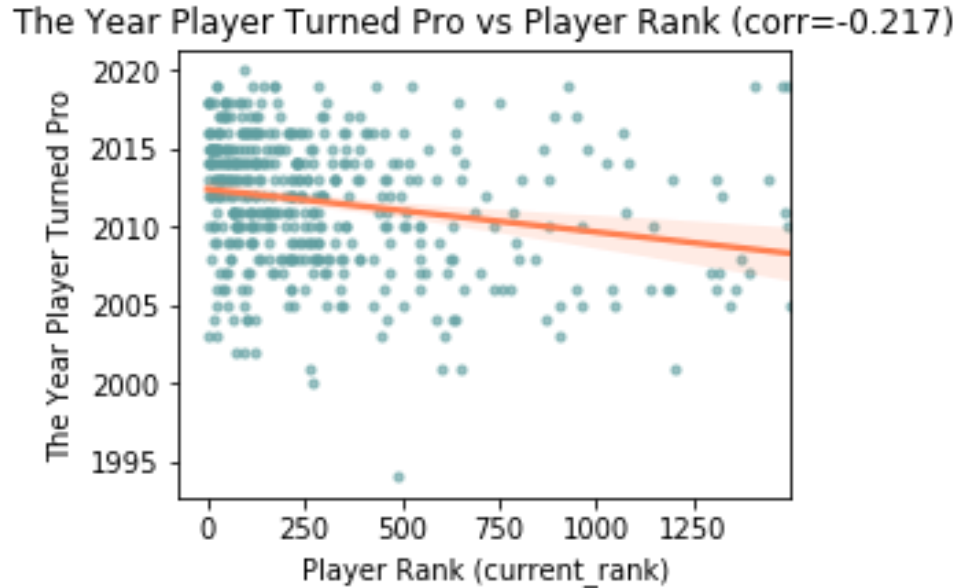
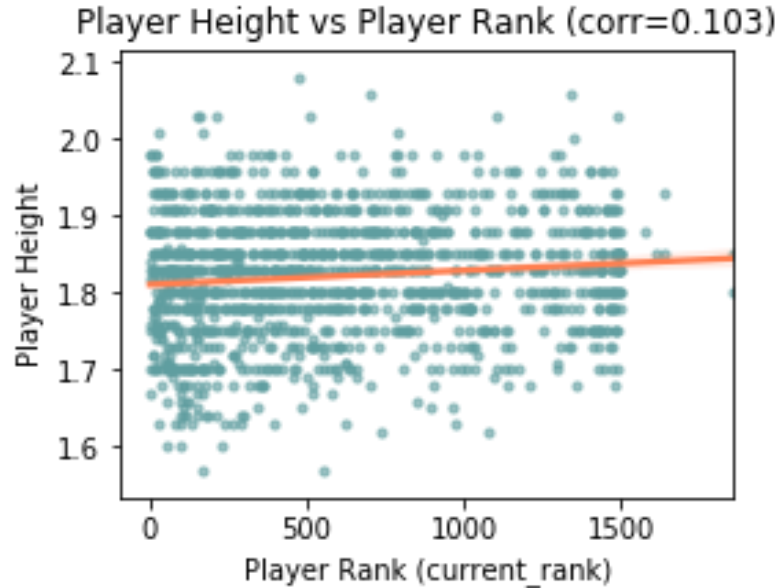
- 
- **Hard Court Outdoor** was the most common surface
  - Players from **the USA** appeared most frequently
  - Players from **France** recorded the most wins

# Match Dynamics



- **Longest** match duration: **4 hours and 59 minutes**
- **Average** match duration: **1 hour and 15 minutes**
- Most matches ended in **2 sets**
- Average **aces** per match: **10.69**
- Average **breaks of serve** per match: **6.92**

# Players Attributes Correlations



- Little to no correlation was observed between height and rank
- Players who turned pro earlier tended to have slightly better rankings. (a weak correlation was observed)

***Thank you for your attention***



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